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10 CFR 50.73

NLS2024035

May 9, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2024-004-00
Cooper Nuclear Station, Docket No. 50-298, DPR-46

Dear Sir or Madam:

The purpose of this correspondence is to forward Licensee Event Report 2024-004-00.

This letter does not contain regulatory commitments.

Sincerely,

Khalil Dia
Site Vice President

/bs

Attachment: Licensee Event Report 2024-004-00

cc: Regional Administrator w/attachment
USNRC - Region IV

NPG Distribution w/attachment

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

INPO Records Center w/attachment
via IRIS entry

Senior Resident Inspector w/attachment
USNRC - CNS

SORC Chairman w/attachment

SRAB Administrator w/attachment

CNS Records w/attachment

COOPER NUCLEAR STATION

72676 648A Ave / P.O. Box 98 / Brownville, NE 68321
<http://www.nppd.com>

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number. EXPIRES: 04/30/2027 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)						
1. Facility Name Cooper Nuclear Station				☒ 050 ☐ 052		2. Docket Number 298		3. Page 1 of 4		
4. Title Main Turbine Stop Valve Position Switches Do Not Meet Channel Independence Criteria Results In Two Channels Being Declared Inoperable And A Condition Prohibited by Technical Specifications										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050
03	11	2024	2024	004	00	05	09	2024	Facility Name	☐ 052
9. Operating Mode 1						10. Power Level 100				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☒ 50.73(a)(2)(vii)				
☐ OTHER (Specify here, in abstract, or NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Linda Dewhirst, Regulatory Affairs and Compliance Manager								Phone Number (Include area code) (402) 825-5416		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS
14. Supplemental Report Expected								15. Expected Submission Date		
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)								
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines) On March 11, 2024, a question was raised concerning the mechanical separation criteria of the Reactor Protection System Turbine Stop Valve (TSV) closure function position switches as the position switches on each TSV are connected together with all-thread for actuation. On March 29, 2024, it was determined that the two position switches for each TSV do not have channel independence and one position switch for each TSV was declared inoperable because the switches are mechanically linked together. The direct cause of the failure to meet the requirement of channel separation comes from the position switches not being mechanically separated due to sharing a common linkage. A causal factor for this was the initial assumption was incorrect that the plant is constructed consistently with the design basis. The channel independence issue will be resolved by the engineering change process. A seminar will be presented to engineering personnel to address the assumption the plant was constructed consistent with the licensing basis. There were no safety consequences as a result of this event.										



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Cooper Nuclear Station	2. DOCKET NUMBER 05000-298	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
		2024	004	00

NARRATIVE

PLANT STATUS

At the time of the event on March 11, 2024, Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 100 percent steady-state power.

BACKGROUND

The Reactor Protection System (RPS) [EIS:JC] initiates a reactor [EIS:RPV] scram when one or more monitored parameters exceed their specified limits to preserve the integrity of the fuel cladding and the reactor coolant pressure boundary and minimize the energy that must be absorbed following a loss of coolant accident. This can be accomplished either automatically or manually.

RPS is arranged as two separately powered trip systems. Each trip system has three logics. Two of the logics are used to produce automatic trip signals. The remaining logic is used for a manual trip signal. Each of the two logics used for automatic trip signals receives input signals from at least one channel for each monitored variable. Thus, two channels are required for each monitored variable to provide independent inputs to the logics of one trip system. At least four channels for each monitored variable are required for the logics of both trip systems.

Closure of the Turbine Stop Valves (TSV) results in the loss of a heat sink that produces reactor pressure, neutron flux, and heat flux transients. A reactor scram is initiated at the start of TSV closure in anticipation of the transients that would result from the closure of these valves.

TSV closure signals are initiated from position switches (also referred to as limit switches) [EIS:33] located on each of the two TSVs. Two independent position switches are associated with each stop valve. Both of the switches from one TSV provide input to RPS trip system A; the two switches from the other TSV provide input to RPS trip system B. Thus, each RPS trip system receives two TSV closure channel inputs from a TSV, each consisting of one position switch assembly with two contacts, each inputting to a relay [EIS:RLY]. The relays provide a parallel logic input to an RPS trip logic channel. The logic for the TSV closure function is such that both TSVs must be closed to produce a scram. Single valve closure will produce a half scram. This function must be enabled at thermal power greater than or equal to 29.5 percent reactor thermal power as measured by turbine supply pressure.

Four channels of TSV closure function, with two channels in each trip system, are required to be operable for Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.3.1.1 to ensure that no single instrument failure will preclude a scram from this function if both TSVs should close.



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EVENT DESCRIPTION

On March 11, 2024, the Nuclear Regulatory Commission (NRC) Senior Resident Inspector (SRI) asked a question regarding the TSV position switches that provide input to RPS. In the CNS Updated Safety Analysis Report (USAR) Chapter VII, Section 2.3.6.4, a statement exists that the switches on each valve are mechanically and electrically separated and satisfy Institute of Electrical and Electronics Engineers standard 279 (IEEE-279). The question from the NRC SRI was since the two switches on each valve are connected together with all-thread for actuation, how does this meet the definition for mechanically separated. Subsequent evaluations were completed on this; however, it was determined on March 29, 2024, that total channel independence was not achieved due to not being mechanically separated. At the time this question was asked, no other TSV closure function position switch was inoperable.

TS LCO 3.3.1.1, Conditions A and B were entered on March 29, 2024, as a result of a required channel in both trip systems for the TSV-Closure function being declared inoperable. A Notice of Enforcement Discretion was approved for this issue on March 29, 2024, to not enforce compliance with the required actions of LCO 3.3.1.1, Conditions A and B. Subsequently on April 5, 2024, a TS license amendment request, Amendment 275, was approved to add a temporary note to LCO 3.3.1.1, Table 3.3.1.1-1, for the TSV closure function, to allow not entering Conditions A and B for this issue until the startup from the next refueling outage.

Review of historical records associated with the TSV closure function design has shown no changes to the method of actuation since original construction. While USAR Chapter VII, Section 2.3.6.4 and Final Safety Analysis Report, Amendment 17, Question 7.23, state that the current design is mechanically and electrically separated and meets IEEE-279, documented correspondence regarding the linkage assembly and its effect on mechanical separation had not been found. Based on the lack of documentation, it is unclear on the level of detail that was provided for the review and acceptance of the original design.

BASIS FOR REPORT

This event is reportable under 10 CFR 50.73(a)(2)(i)(B) – Any operation or condition which was prohibited by TS. This is also reportable under 10 CFR 50.73(a)(2)(vii) as the condition caused two independent channels to become inoperable in a single system designed to shut down the reactor and maintain it in a safe shutdown condition.

SAFETY SIGNIFICANCE

There were no safety consequences as a result of this; however, there is a concern a failure could prevent tripping of both position switches of one TSV, and therefore prevent a reactor scram for the TSV closure function. No instances were found in the last three years that, when combined with the TSV position switches not having channel independence, would have prevented a scram from occurring if both TSVs went closed.



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CAUSE AND CORRECTIVE ACTIONS

The direct cause of the failure to meet the requirement of channel separation comes from the position switches not being mechanically separated due to sharing a common linkage. A causal factor for this was the initial assumption was incorrect that the plant is constructed consistently with the design basis. The channel independence issue will be resolved by the engineering change process. A seminar will be presented to engineering personnel to address the assumption the plant was constructed consistent with the licensing basis.

PREVIOUS EVENTS

A review of CNS licensee event reports (LER) for the last three years was performed. The following reportable conditions are related to RPS TSV closure function:

LER 2024-001-00 – Inoperable Turbine Stop Valve Limit Switch Causes Condition Prohibited by Technical Specifications. [March 4, 2024].